

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 2: Polynomials (Core Basics Edition)

CHAPTER 2: POLYNOMIALS

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

Ye woh cheezein hain jo tujhe pehle se aani chahiye thi.

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

Agle page pe SAARE core basics (RED) ek saath diye hain.

Phir asli Chapter 2, solved examples, aur Q&A.

CORE BASICS - MISS MAT KARNA (RED PAGE)

Ye page sirf un cheezon ka hai jo Class 7-8-9 mein padhayi jaati hain. 10th ka Chapter 2 inhi pe khada hai.

1) VARIABLE AUR CONSTANT

Variable = badalne wala (jaise x , y) - unknown number.

Constant = fixed number (jaise 5, -3, $\frac{1}{2}$).

2) POWER / EXPONENT

$x^2 = x$ times x . $x^3 = x$ times x times x .

x^2 ko "x square", x^3 ko "x cube" bolte hain.

3) POLYNOMIAL KYA HAI?

Polynomial = expression jisme variable ke power sirf WHOLE numbers ho (0,1,2,3...) - negative ya fraction power nahi.

Hai: $2x+3$, x^2-5x+6 , 7

Nahi: $x^{(-1)}+2$, $\sqrt{x}+1$ (in mein galat power)

4) DEGREE

Degree = polynomial ka sabse bada power.

$2x+5 \rightarrow$ degree 1, $x^2-3x+1 \rightarrow$ degree 2.

5) TYPES (degree ke hisab se)

Degree 1 = LINEAR : $ax + b$

Degree 2 = QUADRATIC: $ax^2 + bx + c$

Degree 3 = CUBIC : $ax^3 + bx^2 + cx + d$

(a hamesha $\neq 0$ hona chahiye.)

6) COEFFICIENT

Coefficient = variable ke aage wala number.

$3x^2 - 5x + 7$ mein: $a=3$ (x^2 ka), $b=-5$ (x ka),

$c=7$ (constant). Inhi ko a,b,c bolenge.

7) p(x) NOTATION AUR SUBSTITUTION

$p(x)$ = polynomial ka naam.

$p(2)$ = x ki jagah 2 daal do.

$p(x)=x^2-4 \rightarrow p(2) = 2^2 - 4 = 0$.

8) ZERO / ROOT OF POLYNOMIAL

Zero = woh x jiske liye $p(x) = 0$ ho jaye.

$p(x)=2x-6 \rightarrow 2x-6=0 \rightarrow x=3$. So zero = 3.

(Graph jahan x-axis ko cut kare, wahi zero hota hai.)

9) FACTORISATION (splitting middle term) - recap

ax^2+bx+c ko $(..)(..)$ mein todna.

Middle term b ko 2 numbers p,q mein todo jahan

$p+q = b$ AND $p \times q = a \times c$.

10) GRAPH BASICS

X-axis = leti (horizontal) line, Y-axis = khadi.

Linear ka graph = seedhi line.

Quadratic ka graph = "U" ya ulta "U" (PARABOLA).

Graph jahan x-axis ko touch/cut kare = zero.

NOW: ASLI CHAPTER 2 SHURU

TOPIC 1: GEOMETRICAL MEANING OF ZEROES

Polynomial ka graph x-axis ko jitni baar CUT karta hai, utne uske ZEROES hote hain.

- LINEAR ($ax+b$): graph seedhi line, x-axis ko 1 baar cut $\rightarrow$ 1 zero.
- QUADRATIC (ax^2+bx+c): graph PARABOLA (U shape).
 - * x-axis ko 2 jagah cut $\rightarrow$ 2 zeroes
 - * 1 jagah touch $\rightarrow$ 1 zero (repeated)
 - * cut hi nahi karta $\rightarrow$ 0 real zeroes
- CUBIC: x-axis ko zyada se zyada 3 baar $\rightarrow$ up to 3 zeroes.

[CORE] "Cut karna" = graph aur x-axis ka milna.
Wahi point zero hota hai (RED page point 8 & 10).

TOPIC 2: ZEROES AUR COEFFICIENTS KA RISHTA

EXAM ALERT: Ye relationship aur ispe based questions HAR exam mein aate hain. Formula ratta maar lo.

QUADRATIC $p(x) = ax^2 + bx + c$ ke liye, agar zeroes α aur β hain:

Sum of zeroes: $\alpha + \beta = -b/a$

Product of zeroes: $\alpha \times \beta = c/a$

(Yaad: sum mein $-b/a$, product mein c/a .)

EXAMPLE: $x^2 - 5x + 6$ ($a=1, b=-5, c=6$)

Sum = $-b/a = -(-5)/1 = 5$

Product = $c/a = 6/1 = 6$

(Zeroes 2 aur 3: $2+3=5$ ok, $2 \times 3=6$ ok.)

CUBIC (ax^3+bx^2+cx+d) ke liye (sirf jaan-kaari):

sum = $-b/a$, sum of pairs = c/a , product = $-d/a$.

TOPIC 3: QUADRATIC POLYNOMIAL BANANA (sum & product se)

EXAM ALERT: "Sum aur product diya hai, polynomial banao" - ye bhi baar-baar aata hai. Formula:

Polynomial = $x^2 - (\text{sum})x + (\text{product})$
= $x^2 - (\alpha+\beta)x + (\alpha \times \beta)$

EXAMPLE: sum = 5, product = 6 ho toh
polynomial = $x^2 - 5x + 6$.

SOLVED EXAMPLES: HARDEST -> EASIEST

Ye 8 examples poori tarah solve karke dikhaye hain.

Step-by-step padh - "kaise solve karte hain" samajh aayega.

(Upar HARDEST, neeche jaake EASIEST hote jaate hain.)

Solved Example 1 (HARDEST) - $\alpha^2 + \beta^2$ etc.

(Exam favourite - identities ka use)

α, β zeroes hain $x^2 - 5x + 6$ ke. Nikaalo:

(a) $\alpha^2 + \beta^2$ (b) $1/\alpha + 1/\beta$

Pehle: $\alpha + \beta = 5$, $\alpha \times \beta = 6$.

(a) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2(\alpha \times \beta)$
 $= 5^2 - 2(6) = 25 - 12 = 13$

(b) $1/\alpha + 1/\beta = (\alpha + \beta)/(\alpha \times \beta)$
 $= 5/6$

Solved Example 2 - Ek zero diya ho

(Exam favourite)

$x^2 - 4x + k$ ka ek zero 3 hai. k aur doosra zero?

3 zero hai $\rightarrow p(3)=0$:

$$3^2 - 4(3) + k = 0 \rightarrow 9 - 12 + k = 0 \rightarrow k = 3$$

Product of zeroes = $c/a = k/1 = 3$.

$$3 \times (\text{doosra}) = 3 \rightarrow \text{doosra zero} = 1.$$

(Check sum: $3+1 = 4 = -b/a = 4$. ok)

Solved Example 3 - Zeroes nikaal + verify rishta

(Exam favourite - factorisation + relationship)

$6x^2 - 7x - 3$ ke zeroes nikaalo aur rishta verify karo.

Split middle term: $a \times c = 6 \times (-3) = -18$,

do numbers jinka sum -7, product -18 $\rightarrow -9$ aur $+2$.

$$6x^2 - 9x + 2x - 3$$

$$3x(2x - 3) + 1(2x - 3)$$

$$(2x - 3)(3x + 1)$$

Zeroes: $2x-3=0 \rightarrow x=3/2$; $3x+1=0 \rightarrow x=-1/3$

Verify:

$$\text{Sum} = 3/2 + (-1/3) = 9/6 - 2/6 = 7/6 = -b/a = 7/6 \text{ (ok)}$$

$$\text{Product} = (3/2)(-1/3) = -1/2 = c/a = -3/6 = -1/2 \text{ (ok)}$$

Solved Example 4 - Polynomial banao (zeroes diye)

(Exam favourite)

Zeroes 3 aur -2 wala quadratic banao.

$$\text{Sum} = 3 + (-2) = 1$$

$$\text{Product} = 3 \times (-2) = -6$$

$$\text{Polynomial} = x^2 - (\text{sum})x + \text{product}$$

$$= x^2 - (1)x + (-6)$$

$$= x^2 - x - 6$$

Solved Example 5 - Polynomial banao (sum/product diye)

Sum = -3, Product = 2 wala quadratic.

$$\text{Polynomial} = x^2 - (\text{sum})x + \text{product}$$

$$= x^2 - (-3)x + 2$$

$$= x^2 + 3x + 2$$

Solved Example 6 - Graph se zeroes count

Graph dekh ke zeroes batao:

- x-axis ko 2 jagah cut kare -> 2 zeroes

- 1 jagah sirf touch kare -> 1 zero (repeated)

- x-axis ko chhuye hi nahi -> 0 real zeroes

(Number of zeroes = number of cutting points.)

Solved Example 7 - Verify rishta (simple)

$x^2 - 2x - 8$ ke zeroes aur rishta.

Factor: $(x - 4)(x + 2)$ -> zeroes 4 aur -2.

$$\text{Sum} = 4 + (-2) = 2 = -b/a = 2 \text{ (ok)}$$

$$\text{Product} = 4 \times (-2) = -8 = c/a = -8 \text{ (ok)}$$

Solved Example 8 (EASIEST) - Linear zero & p(2)

(a) $2x - 6$ ka zero: $2x - 6 = 0$ -> $x = 3$.

(b) $p(x) = x^2 - 4$ -> $p(2) = 2^2 - 4 = 0$.

(Matlab 2 bhi ek zero hai is polynomial ka.)

Q AND A TIME

(Pehle khud try kar, phir mujhe bhej - main check karunga)

Q1. $p(x) = 3x^2 + 5x - 2$ ka degree aur type bata.

Q2. $p(x) = x^2 - 4$ ke liye $p(2)$ aur $p(-2)$ nikaal.

Q3. $x^2 - 7x + 12$ ke zeroes nikaal aur sum/product rishta verify kar.

Q4. Zeroes 5 aur -3 wala quadratic polynomial banao.

Q5. Sum = 4, Product = -5 wala quadratic banao.

Q6. α , β zeroes hain $x^2 - 6x + 8$ ke.
 $\alpha^2 + \beta^2$ nikaal.

Q7. CORE CHECK: $x^{-1} + 3$ polynomial hai ya nahi? Kyun?

Q8. CORE CHECK: $2x^2 + 3x - 1$ mein a, b, c kya hain?

SUMMARY

- Zeroes = jahan $p(x) = 0$ (graph x-axis ko cut kare).
- Number of zeroes = graph x-axis ko jitni baar cut kare (linear 1, quadratic up to 2, cubic up to 3).
- Quadratic ax^2+bx+c ke liye:
Sum of zeroes = $-b/a$
Product of zeroes = c/a
- Polynomial banao: $x^2 - (\text{sum})x + (\text{product})$.
- Useful identity: $\alpha^2 + \beta^2$
 $= (\alpha+\beta)^2 - 2(\alpha \times \beta)$.

CORE BASICS (RED page) revise karna - polynomial, degree, coefficient (a,b,c), zero, $p(x)$, parabola.